A STARTUP BURNS $50K/MO ON GPT-5.

80% OF THEIR QUERIES ARE SIMPLE FAQS.

"That's an engineer's salary. Up in smoke."

For what? A question a 1.5B model could answer in 300ms.

THE PROBLEM

THE AI INDUSTRY HAS A ROUTING PROBLEM.

30+

NEW MODELS PER MONTH

No team can evaluate them all. So everyone picks one and never revisits.

50–90%

WASTED INFERENCE SPEND

Enterprise budgets burned on over-provisioned models. Money that could hire engineers.

180×

ENERGY PER PREMIUM QUERY

39 Wh vs 0.22 Wh. AI on track for 1,000 TWh by 2026 — mostly waste.

WHO IT HURTS

Developers	10-30x OVERPAY
Enterprises	COMPLIANCE RISK
End Users	SLOW RESPONSES
The Planet	415 TWh / YR

YOUR AI IS PAYING THE TOLL.
For nothing.

THE FIX

ONE LINE OF CODE.
We handle everything else.

BEFORE

STATUS QUO

```
# Same premium model. Every request. Forever. from openai import
OpenAI client = OpenAI( base_url="https://api.openai.com/v1")
response = client.chat.completions.create( model="gpt-5-4", # ←
HARDCODED messages=messages)
```

AFTER MODELROUTER

DROP-IN

```
# Right model. Every request. Automatically. from openai import
OpenAI client = OpenAI( base_url="https://router.acme/v1") # ← ONE
CHANGE response = client.chat.completions.create( model="auto", # ←
ModelRouter decides messages=messages)
```

API COMPATIBLE

Full OpenAI spec. Zero rewrites. Drop-in replacement.

ALWAYS OPTIMAL

Cheapest capable model per request. Automatically.

CONTRACT-AWARE

Upload SLA docs → routing rules extracted. Per request.

ARCHITECTURE

TWO PHASES. FULLY AUTOMATED.
Zero ongoing effort.

PHASE 1 — ONBOARDING = 30s

- **Upload Contract**
  PDF or text — SLA, privacy, custom rules
- **LLM Extraction**
  Frontier model reads and parses all constraints
- **Customer AI Profile**
  Region, providers, latency target, routing tiers
- Endpoint Live
  /{customer}/v1/chat/completions — ready

PHASE 2 — RUNTIME = 50ms

PROMPT IN → CLASSIFY → FILTER → SCORE → ROUTE → RESPONSE

SIMPLE	GPT-5.4 nano, Gemini Flash Lite, Haiku 4.5	~300ms
MEDIUM	GPT-5.4 mini, Sonnet 4.6, Grok 4.1 Fast	~700ms
COMPLEX	GPT-5.4, Opus 4.6, Gemini 3.1 Pro	~1.5s

Classifier: Arch-Router-1.5B · 62ms on GPU · heuristic fallback

FINE-TUNED 1.5B ROUTER — 62MS CLASSIFICATION

TRAINING PIPELINE

Qwen/Qwen2.5-1.5B-Instruct — Base

↓ Katanemo

katanemo/Arch-Router-1.5B — Intent router

↓ GRPO FINE-TUNE ← OUR WORK

ModelRouter-Router

↓ GGUF Q8_0

ModelRouter.Q8_0.gguf — 1.6 GB, production

Method **GRPO + Unsloth**	LoRA Rank **32 (2.3%)**
Training **150 steps**	Time **2.5 min**
Hardware **RTX 3080 8GB**	Dataset **172 prompts**

BENCHMARKS

SIMPLE	MEDIUM	COMPLEX
81.8%	**85.7%**	**85.7%**
Stock: 87.9%	Stock: 14.3% +71.4pp	Stock: 100%

Overall Accuracy **83.3%**

TECH STACK

FastAPI · Next.js · Arch-Router-1.5B · OpenRouter · GRPO · Docker · SQLite · GGUF · Python · TypeScript

Cost Savings vs Premium **59%**

IMPACT

ONE CHANGE. BENEFITS AT EVERY LAYER.

LAYER	BENEFIT	HEADLINE
BUSINESS	Eliminates 60–98% of AI inference spend through intelligent right-sizing	**60–98% SAVED**
USERS	Simple queries → fast answers. Complex queries → powerful models.	**RIGHT MODEL**
DATA CENTERS	Less unnecessary load on premium GPU clusters. Resources freed for real work.	**LESS WASTE**
ENVIRONMENT	Up to 180× less energy per routed-away premium call. Thousands of calls saved.	**180× LESS**
COMPLIANCE	Contract constraints → routing rules. Enforced per request, automatically.	**ZERO ERRORS**

LIVE DEMO

FIVE STEPS. ONE CONTRACT. IMMEDIATE ROUTING.

01
Onboard ACME

Region: EU-only from GDPR
DeepSeek blocked
Latency: 1000ms

02
Copy Endpoint

/acme/v1/chat/completions
Drop into any OpenAI client
Zero other code changes

03
Simple Query

"What is your return policy?"
→ SIMPLE
→ GPT-5.4 nano · 300ms

04
Complex Query

"Analyze liability exposure..."
→ COMPLEX
→ Opus 4.6 · Same endpoint

05 — Dashboard

Model distribution · Cost savings 59% · Live request feed · Per-request explanations

LIVE

THANK YOU.

TEAM DYNAMIC IP

BACKEND & AI
Vardhan

FRONTEND & PLATFORM
Rajesh

"One URL change. Every model. Optimal choice. Zero code changes."

HackArena 2.0 | Generative & Agentic AI | ModelRouter

github.com/dynamicip/modelrouter

FUTURE SCOPE

✓ MVP delivered ○ Vector search ○ Provider health ○ Managed SaaS